

ActInf Textbook Group ~ Cohort 1 ~ Meeting 2 (Chapter 1)

TextbookGroup/ParrPezzuloFriston2022/Cohort_1 | Cohort 1 ~ Meeting 2 (Chapter 1) | 57:35

hello everyone thanks for joining the
seconds
meeting
where we're going to be discussing
chapter one
it's the active textbook group and it's
may 12 2022.
you can click on my screen and that will
show you the full screen
we updated the live meeting page a bit
and it's
just like all these other pages a work
in progress but hopefully this will be a
structure where
we'll be able to hear everyone's
perspective and leave the coda in a
better place than we found it
so what we'll do in the live meetings
which from here on out
or at least for here for the next two
and a half months or so are going to be
like
driven by the material
in the chapters and in the appendices
we're going to go to the questions page
and we're gonna look at the questions
that have been asked for that week for
that theme
and we're just gonna start with the most
upvoted questions and then we're gonna
be taking notes
in the answers in the discourse and
we'll go into a little bit more detail
when we flip to that page
and then after addressing questions for

probably most of the time and discussing
that
we'll go
pull back a little bit more broadly like
ideas
for that
week um
and then in the last minutes we can just
see if there's been any changes to the
project ideas
and start to see what people are excited
about applying active inference to
and then we'll end at the top of the
hour
and take a quick break and then there'll
be a tools organizational unit meeting
following for those who want to just
like listen or participate in dot tools
so let's head over to questions and i'm
going to hide the sidebar because we're
going to be mostly here in questions
okay
so it's awesome
and also control and plus minus or
control and scroll is like a good way to
resize how it looks
um
so this is our first week with this
format
and anyone's welcome to like
suggest a a variant or a different way
to carry this out but
um the co-organizers of the textbook
group kind of
thought we could
do something along these lines
the questions we're gonna be looking at
are in chapter one

so that's the chapter we're discussing
today
and we're going to start with the ones
that have the most upvotes in the
interesting
um
and just
spend a couple minutes on each of the
questions
and at any point
there are the affordances of just
listening
and also of course re-watching this
recording
there's also the affordance in gather to
raise your hand if you'd like to share
on that question that we're going to
have up
and additionally
in the um by clicking on the answers in
discourse
that will pull up like a focused view on
that question
and everybody can collaboratively edit
the answers in discourse and so it's
better to like add and be noisy even if
it's a bullet point list or an
incomplete sentence or there's
incoherence or it's contradictory those
are all fine
it just like adds information and
perspectives
to the question so yeah we're going to
go to each question
open up
the answers in discourse
read the question
see what has already been contributed in

the answers in discourse and then also
like if people want to share
other thoughts that they're having
that's awesome
and then also if anybody wants to be
like taking other notes just things that
are coming to mind that they don't want
to share
in a speaking way
or just taking some notes on what other
people are saying
um
so
let's just
go into it and i'll be just keeping an
eye for the raised hand in gather
or anybody who wants to speak so
um starting with this
first question
maybe we can have like a like addressed
during recording or something like that
but
so
how do you understand the distinction
between the high road and the low road
to active inference
what do you think is interesting or
useful about this distinction
so does anybody want to like just
summarize how they
were thinking about the high in the low
road or we can look at what um others
have already written here
what is the difference between the high
road and the low road
are there other roads
are there high and low roads to other
theories

what is this metaphor with the roads
so let's check out yes um ali and then
mallory
well i believe it's just um another term
for top-down or bottom-up approach in a
kind of way
and um well the
high road i believe starts uh from
the actualized mind itself and tries to
derive
the underlying mechanisms or underlying
principles
but the bottom-up approach or the
uh
in other words low road
uh tries to
start from
the elements
tries to start from the
unorganized elements let's say and
derive the higher mechanisms or higher
order mechanisms
actualizing the mind itself
thank you awesome allery
um yeah i was just gonna
maybe recapitulate i guess what i wrote
in the comments and
i see now that i probably should have
written straighten the answers in
discourse section so sorry about that um
but um
i really appreciated this
schematic
because
um
i've read
papers that i think come at active
inference from both of these angles and

in trying to understand
i don't know the framework i think
um
it wasn't super clear to me how all of
these different terms that are like
really nicely outlined in this figure
um figure 1.2
um
are related to one another so just to
have it
i think
laid out like this it was really very
very helpful um
and i think um
still being kind of new to the concepts
and and such i liked the way they framed
it in terms of kind of
um the high road describing
um
the kind of what and why
and the low road being more about the
how
um
i thought that was really
um
very useful
so yeah i really i love this and i i
saved it and we'll probably reference it
quite often
yeah agreed it's it's really uh useful
and it touches on
aristotle's multiple y's
where some of the y's are are more like
hows
um or they're sort of proximate
mechanisms or more ultimate mechanisms
um and and we can like look at figure
1.2 so active inference is in the middle

and the high road i guess this is the
starting train stop here with a free
energy principle
um which
some later questions also might invoke
and then it kind of goes down to active
inference whereas starting with bayes
theorem
also gets us to
sentience
behavior
sentient being like sensing and feeling
and acting not
as much on the um
qualia and experiential side
um
so what else do we
have in the answers so high road what
and why
that is
starting with sort of the organismal
imperative or or like um ali i think
said there that the actualized mind and
then what are the underlying principles
that make
adaptive behavior adaptive behavioral
systems
working
down you know the castle in the sky is
the adaptive behavior and intelligent
cognitive behavior
what
is under it
and then the low road
doesn't begin with that imperative it
starts with like the elements
like the low road starting with base
none of these necessarily have to be or

at least maybe until you get to like
bayesian brain
these don't have to be adaptive
a generative model doesn't have to be
resisting entropy
so it's sort of like here's where we see
a lot of the elements in the machinery
and then on the top is more of the
self-organization
resistance to dissipation autophagocytosis
how are you going to get there you have
to be minimizing your surprise
what does that actually
entail and then active inference is like
this intermediate stop
and again maybe there's other roads or
there's other paths
um
yeah
so any other thoughts on high road and
low road or we'll continue to the next
question
this is just being
introduced in chapter one
because the following chapters two and
three which we're going to get to after
the appendix a and b
are going to detail the low road and the
high road
so this is kind of like a map for some
of the next steps that we're going to be
engaging in
okay
next question
the concept
of organisms minimizing surprise is
introduced on page six
how are prior assumptions or beliefs

established when the organism is in new
or novel environments

so anyone can raise their hand or
take a note here

um

mallory or ali with raisins

one helpful approach might be to think
about like the questions asking about
new and novel environments so are we
dealing with

um categorical novelty

or with like quantitative novelty like i
was measuring the temperature and then

i'm in a new environment where the
temperature is different

that's one type of novelty but there

might also be like this like

novelty

at an even higher level with like i was
looking at

um

previously i was looking at temperature
and now i'm in a place where i'm having
priors about smell

so that's like a sort of intermodal

or like

a different

type of thing that's being tracked so

that's a categorical novelty

as opposed to

just updating the temperature parameter

in a different temperature room

whether

the person who asked it is here or not

um oh yes okay ali and mike go for it

uh yeah um well again if we're talking

about um

the um human mind

uh
i mean uh if
uh
from the priors
we can infer that uh well it's something
that's
kind of innate in human minds is
something uh wired and
i mean
perhaps genetically or uh in any other
biologically
hardwired way but
well
again yes the definition of novel
situation varies uh
according to
what we're exactly referring to
and if uh a novel situation is something
that the organism uh hasn't
uh ever presented with uh well in that
case uh these prior
these priors or these innate tendencies
won't
help it very much so in that case
it should probably go
along the
way of
constructing
posteriors more actively i believe than
in any regular situation
thank you ali
uh jessica
and then mike and then steve
hi um i think
maybe like in this situation even though
the um
the agent might be encountering
something that has an experience before

they might use something as proxy that
they already know
and sort of like project
those understanding into
that new situation and then it's through
action etc then they're gonna be
updating like okay how true is it or
isn't
to what they understand but they might
be making some kind of projection based
on what they already know and things
that look similar to that new experience
awesome thank you uh mike and stephen
yeah i was just gonna comment um
in the case where the organism is brand
new and has no experience um
i think ali's comments cover this a bit
that it will have to be updating its
priors fairly rapidly and
presumably starting with sort of a wider
distribution
great stephen
yes along the lines of what jessica was
saying
there
has been some work in systematizing that
as case-based reasoning
not necessarily in a
bayesian framework here
but as a
derivative of expert systems
where
when you get a
new scenario you try to match that
against previous scenarios that you've
already seen and adapted
however necessary to
meet the new observations

so that might be a place to look for
uh inspiration
thanks
yeah and in in the
ontology work and in a lot of these
questions there's often like a narrow
answer and then a broader answer like
the narrow answer would be well again if
all your bayesian model is doing
is looking at the thermometer reading
and then doing a column and filter and
inferring the temperature in the room
well then the answer of what happens
when you put it under water or what
happens when you ask it to predict scent
it's kind of like a trivial answer which
is it's just not a parameter in that
script
nothing happens
and then that's like the narrower answer
which is
often very deflationary or like very
trivial
answers
and then there's also a broader question
like in the ecological context or in the
organismal context in the expert applied
context
what happens when there are these
different qualitative and quantitative
types of novelty
and so sometimes it's like
um
there we are in the middle again with
the deflationary or the narrow being too
obvious and too simple to be brought
into action
and then the bigger question like how do

experts

in medicine or

some other field how do they handle

different types of novelty is like

that's its own

research question and agenda

awesome

any other thoughts on this question

about

priors and it's also very important note

that surprise is always relative to

expectations

which are priors in the bayesian context

and those are granted by the generative

model so surprise is not just free

floating it's in relationship to the

outcomes as expected by an organism for

example jessica your hands raised

otherwise we'll go to the next question

awesome

okay this question i think is pretty

similar to the previous

um one so if anybody wants to give a

thought otherwise no worries

are the high road low road the only

roads to active inference or my active

be like rome

which i guess we can unpack a little bit

many slash all roads leading to it

um jf

yes that's more of a general question uh

uh

if for example if what you're trying to

understand is race cars right and you

develop a mathematical framework a given

number of measurements like the the

weight of the car the length of the car

the torque of the engine the adhesion of

the tires the friction and whatnot
you're able to predict how well the the
race car is going to perform in
different conditions and whatnot
does that mean that you're an expert in
race cars does that explain the race car
could you engineer a race car
based on this knowledge so
talking about multiple the different
roads um
one road through the free energy
principle i think is an analytical
uh road
that looks at
the outcomes
of the
the machinery of the uh the living
beings that try to maintain homeostasis
but how much does it say about the
implementation
of those
organisms
it says
many things but does it say everything
that we deem
might deem interesting
so what are the limitations of of this
this this uh
framework the the
the the active inference framework uh
defined by the free energy principle
what are its limitations
on top of what it the
given its capabilities which are which
are i think uh enormous but there are
limitations and and if there are
limitations that means there are there
must be other roads

that
can be taken
to
circumvent these limitations
thank you great comment ali
uh well
delusion gathering in a thousand
plateaus
has
laid out um the concept of rhizome i
don't know if you're familiar with the
concept of rhizome as opposed to
an arborescence structure
well
basically the concept of rhizome
uh expounds a
structure and now a knowledge structure
in which every concept can potentially
be connected to every other concept
without having to go through a principle
uh trunk
of a knowledge struct a knowledge tree
let's say but uh in an arborescent
structure and not arborescens knowledge
structure
everything branches off from a principal
trunk from a principle
uh let's say
uh
and i believe that in active imprints uh
a rhizomatic approach can also be
uh
explored
and
i mean
not only by
uh
approaching uh

to different concepts via the uh
high road or the low road
main or principal trunk
it can potentially be approached from
uh seemingly unrelated concepts too
awesome like to
maybe even approach a limitation or a
seemingly unrelated concept
both the high and the low road in this
textbook
are formal and they're both grounded in
like mathematical symbols
so are there qualitative
or
non-mathematical
roads
maybe
some of the early roads are mathematical
but
is it possible that a future
road or path or tunnel or
choose your travel metaphor might not be
that case
and then and then thanks for sharing
that um the resource there
because um this is another interesting
angle
which um
[Music]
various people have traced this
evolution of visual metaphor
where it's like things are ladder-like
and scalar
and then there's more of a bifurcating
tree
and then that becomes
um critiqued or complexified into more
of a network

like

with introgression among species for
example so there's sort of this movement
from

linear

through a tree like a directed acyclic
graph towards something that's more
networked

so

that's um a lot of interesting
ideas anything else that people want to
add or we'll continue on

okay

on page 10 they write creatures engaging
in active inference actively seek out
salient sensory observations that
resolve their uncertainty

so

what is the relationship between
salience in the kind of observation that
reduces uncertainty

why is the relationship between salience
observation and uncertainty this way
how does active inference frame this
relationship similarly or different from
which other specific frameworks

so someone wrote

i like this question a lot eager to get
a better understanding of what
constitutes salience

is salience referencing information that
is self-evidence or information that is
precise and surprising or dot dot dot
yes i think um

attention and salience

is addressed more in papers on active
inference

than at least just from a preliminary

pass of the book so attention and
metacognitive mechanisms are not going
to be
um
a core
feature

in the book however there's again been a
lot of interesting work on like the
neurobiology and the
computational aspects of salience and
attention

we can click on the word and and see
a definition but this is not like the
final definition this is just what
some have gotten to salience is the
extent to which a cue or an observation
commands the attention of an agent given
their regime of attention so potentially
a little bit taught logical
like a salient

observation is something that is
grabbing attention a salient cue is
something that commands attention steven
[Music]

yes i would say that especially if we're
defining it in
that way that there are
two elements to salience and one is what
was

pointed out in the question i guess that
you are reducing
uncertainty

but there is also the question of what
uncertainty is worth reducing
and that gets into
utility and where

where it makes sense to explore
i don't know if we want to

choose to change the definition of
salience to just be the second of those
or leave it as having both of those
elements

thanks

mallory

um i think something i don't necessarily
understand um which i would appreciate
maybe somebody speaking to a little bit
is um

this is related actually to a question i
put in earlier about the distinction
between

um

perception and learning but um
something that grabs attention
are we distinguishing so that would say
that something is
like the point that was just made
something is salient if it reduces
uncertainty

um if we're also

attending to things in order to actively
construct a niche that is consistent
with

um

our

model of the world

would

would we not consider that
something

like that object as attracting attention
as opposed to art like how do we
distinguish a guess between active
sampling and attending

um

and how is that related to salient so
something salient either if it

um
is
surprising
or
if it is
like um
serving the
purpose of kind of like fulfilling a
prediction i i'm not sure i understand
that kind of distinction i'm not sure
that makes sense but
awesome question
and
this is one reason why
looking at the formalisms and how
they're framed will help us because in
the purely
english linguistic realm
we could
imagine sentences
of any
type
so in the active inference formalisms
what what are being tagged with salience
which variables are said to reflect
salience and then are they
um or in what ways are they
related to this concept of surprise
which we're going to get to in a
question soon
versus
utility
and this is also speaking to
the way that expectations and
preferences
in active inference are framed
differently than in certain other
frameworks

um can i jump in and just ask another question about this question rather than an answer which is um wouldn't some data be salient if it's even if it's sensible like we might have two levels of salience so i just put this quote in about the notion of a umwelt being a more or less broad series of elements called carries of significance or marks which are the only things that interest the animal now interest the animal is one thing but even just perceiving those marks as existing i mean active inference does have room for that in terms of each agent having some sensory data so you could say well since i can't see uh infrared light it's not salient to me when i'm out in a field of flowers but a bee can see that i think it's you can see all infrared or ultraviolet i came over which one um so you know it may be a salient concept for me as a thinking agent but it's not one that's salient for me when i'm navigating a field of flowers basically nice thanks for bringing in um umwelt and a lot of ecological psychology terms do make their way to active and so then you brought up like levels of perception and salience for example there's things that are outside of the wavelength that can be perceived so photons are not going to be perceived at that wavelength but then there might be within the

visible range

novel

arrangements

like there's something moving there

or there might be something that is

static yet commands attention like the

mona lisa

so what is happening when somebody's

regime of attention is fixated on a

static artifact

versus

something where they're being fixated on

something that's rapidly changing like a

movie

mallory in the chat you wrote can folks

recommend papers so definitely people

can recommend papers

anywhere they see fit

and one place to check out is in the

bottom of our coda in the live streams

under the live stream tab

you'll find like

45 um papers that we've relatively

extensively discussed

and so these are are not like the only

or the canonical papers or anything but

there's a lot of important ones in there

so

especially if you're interested in

looking at papers where like people in

the lab might have familiarity or

there's some things to listen to or

watch about it

then these are some good places to start

all right

awesome

next question

all right this is a contextualization

question

this is not a metadata contextualizing
sentence

on page 9 and 10 they talk about
differences from alternative frameworks
for example in contrast with other
frameworks

policy selection in active inference
automatically balances exploration and
exploitation

and later active inference extends the
inferential approach to domains of
cognition that are rarely considered
what are

five to seven examples

names in a sentence of other theories
and frameworks that active inference
improves upon or updates

anyone can raise their hand this is an
excellent um

homework

flash collaborative stigma g question

because even if each person only thought
of one novel

answer here

with a link and a name or somebody
improved what somebody else had already
brought

very quickly

as a textbook group we could have like
quite a coherent and comprehensive list
the idea of balancing exploration and
exploitation

so called is going to be

a key

theme

and the way that active inference
does

handle

this trade-off

not through the use of like an
exploratory mode or exploitative mode or
some parameter that is like governing
the the handoff there like some sort of
you know

x times exploration

plus one minus x of exploitation

that is like one approach that's
commonly used but

it would be

um a question of like where you know
which citations use that type of an
approach

and it's

going to come

to the epistemic and pragmatic value
whereas

um when there's little epistemic value
to be gleaned little informational
uncertainty result to resolve
then pragmatic value can be pursued
which is

reducing the surprise about
outcomes in relationship to preferences
and then on the other hand when there's
a lot of uncertainty

exploratory behavior might be engaged in
because there isn't enough
certainty around how to pursue those
pragmatic aims

and these are going to come into play
with like the team

mouse model that's going to be later in
the book

but this is a great question so anyone
want to like just give an example of

some other theory or framework
or some other approach to thinking about
exploration exploitation
one starting reading if anyone is
interested there is a 2015 paper
that is looking at exploration and
exploitation trade-offs
in different domains ali
well i don't know if it necessarily
improves upon but
certainly an alternative approach to
this theory
can be
stephen grossberg's art theory the
adaptive resonance theory that we talked
about a previous session
in that theory
uh the problem of learning and cognition
uh is attacked from the viewpoint of
solving
the stability slash plasticity dilemma
and uh in this context in the context of
art or adaptive resonance theory uh well
uh
or
let's say the difference between the
active imprint one of the main
differences between the active imprints
or any
predictive coding schemes
from
adaptive resonance theory is that
in bayesian models
generally
the fully
fully
suppressive matching
circuits uh i mean uh the uh

[Music]

the circuits that can propagate the
information it cannot solve the
stability slash plasticity dilemma as
opposed to the art or adaptive resonance
theory

so plasticity being more exploratory and
openness to updates

and stability being more like parametric
lock-in

more exploitative within a certain um

well yeah the plasticity refers to a

system that doesn't have

uh catastrophic

catastrophic forgetfulness

awesome

cool so we can just you know have this

question up and check back and people

can add things we'll just look at a few

more questions

so

on page 8 they write

the high road perspective is useful to

understand the normative nature of

active inference

what the living organisms must do to

face their fundamental existential

challenges

minimize free energy

and why the what and the why in the high

road

why to vicariously minimize the surprise

of their sensory observations

what does it mean to minimize surprise

vicariously

so someone has written

interesting choice of words here are two

possible answers also the authors or

the readers participants may have
something else in mind
vicarious might refer to the point on
page six that agents cannot tractably
minimize surprise as the authors
technically defined it
in chapter two coming up
so instead they're going to minimize an
approximation to surprise which is
variational free energy
however i'm not sure if they're
referring to
the two
types of surprise that are going to be
approached in chapter two which is the
marginal surprise
um like informational surprise and then
the bayesian surprise
or both
so one option is
surprise as defined
one way or the other or both
cannot be directly
calculated or it's not tractable to to
pursue and so the vicarious could mean
like indirect or via a proxy
and in that case something like
variational free energy is being
vicariously minimized and then that is
going to be used to
get at the surprise minimization and the
second option the person has written
vicarious might refer to the point that
agents minimize free energy not only in
response to real sensory observations
but in anticipation of them
those anticipations could come from
internal simulations by subroutines or

other modules or memory from recall of
past observed events

page 32 which is probably in chapter two

the authors seem to be referring to

engaging models vicariously in a counter

factual sense dealing with imagined

surprise at presumed observations

so these are really important comments

we have the indirect minimization of

surprise

using a proxy

so some type of heuristic to minimizing

surprise now

and so that is vicarious but

contemporary

and then the second sense

is

whether we're using a proxy or not

is referring to

um

[Music]

observations that haven't occurred so

anticipated or expected observations

some of these hopefully we'll come back

to in chapter two

but these are important questions and

it's gonna speak to

like variational free energy

and expected free energy and free energy

of the expected future a few other

constructs

stephen and yeah and then

yeah i put that last comment and i was

just thinking that should probably go

under number two there so i guess that

would be

evidence of that second interpretation

thank you

okay

so we'll maybe return to this in um

chapter two

okay let's just see if we can look up

just a few of these questions it's

awesome like how many there are and of

course anyone can always

read and contribute asynchronously

all right

the subtitle of the book is the free

energy principle in mind brain and

behavior what is the relevant

distinction

between mind and brain in the context of

the free energy principle

someone has written really appreciate

how the researchers have started

proposing biological implementations of

active in the brain

but it's not clear to me how this maps

onto related psychological concepts like

colloquial surprise predictions salience

arousal

yeah awesome and there's been some

recent work and some work that we're

going to be looking at in um

live stream 46

on active inference and folk psychology

about connecting some of the formalisms

to folk psychological concepts so

that's one

place we'll be

looking at that but

um

[Music]

how does

or does active approach this mind brain

question

whether the cartesian

duelist question

hard problem of consciousness

intentional stance of denrit

why is the subtitle of the book that way

jessica and then anyone else with their
hand

um yes i mean i don't know to the answer

but

i

i guess make a guess i think maybe they

um mind will be more the cognition and

the brain looking at the physiology

and

and then you know looking at behaviors

that would be my guess

yeah awesome

bacterium does not have a brain

but does it have a mind

also

great questions we'll just continue just

these

questions with fewer upvotes because

there's such a bunch just so we can get
them in

so

generative model is used in the singular

or

is used in the singular as in the

generative model

or in the plural as in hierarchical

generative models

is a generative model composed of

generative models i.e is it structurally

fractal

someone has written personally i think

the answer is yes

or it's possible to answer yes

via referencing a paper that talks about
blankets of blankets markov blankets
life this paper definitely
does um think about the nesting of
multi-scale complex systems
in terms of like nested generative
models
that then
are
constituting
yes
that are then constituting um generative
models at a higher level
so like there's people with their
generative model and then the team has a
generative model of a different type jf
yes but even in individual
myself yourself
um
it can be could be characterized as
having a generative model
composed of uh
other general models dealing with uh
less abstract uh umwelt for example
so i i think i think yes i think almost
trivially in the sense of a collective
but in the sense of an individual uh
does an individual also
can be said to possess a generative
model that is itself composed of
sub-generative models
awesome point and we've
in several
live streams and settings discussed like
there's one sense where nesting is used
to refer to systems that are like
potentially even physically nested
and then we also see the nesting

architecture

for example here it's live stream number

25

when we discussed it but this is a

nesting

of

attentional and cognitive models

so this is even of a so-called

individual so what is it

about model nesting when sometimes it

can refer to

physical enclosure

and other times it is something that

enables counterfactual reasoning

attention and metacognition

so what is the nesting and and if

someone is just seeing this kind of a

notation or this kind of a graph for the

first time we're gonna totally

go into it but it's gonna be one of the

core formalisms and also this paper is

going to deal with nested models and

other papers do the textbook is going to

spend almost all the time

um on just one level so just sort of the

kernel or the essence or the archetype

or the essential motif

of this partially observable markov

decision process

and the textbook is going to focus on

juxtaposing the discrete time

partially observable markov decision

process with a continuous time

but it will deal less with hierarchical

nesting but it does bring it up but it's

not going to be focused on

okay

what is the information content of a

prediction

i'm just going to read these ones just
so that those who are listening can see
like the kinds of questions that they
could be adding or addressing
this question has to do with the
relationship between active inference
and pleasant surprises
potentially even love languages
receiving money on the pavement
page five

is uh speaking to the author's rhetoric
around theory proliferation and
integration where basically they are
saying well we could have a dedicated
explanation for every neurotransmitter
for every synaptic junction every unique
behavior every cognitive phenomena
or we could somehow utilize first
principles whatever those are to result
in theory integration
and that gets mapped onto the scruffies
and the neats
is it really helpful to redefine terms
such as probability and surprise to
sometimes mean utility
and diversions from utility if that's
what the authors are doing there appears
to be an unhealthy ambiguity in how they
use these terms
this is where the formalisms are gonna
really come into play
um and also the um
[Music]
the live stream on
free energy a user's guide

37

deals with this um dual interpretation

of expectations and preferences and this is like one of the most essential differences between active inference and reward learning or reinforcement learning frameworks is a dual deployment of the same variable in formalisms to represent preferred states as well as expected states like homeostatic range is expected and preferred it's where one wants to be found and it's where one expects to be found and so one can realize those preferences by reducing their surprise about the observations rather than saying well homeostatic states are valuable and we're going to pursue maximum value

i might it be how might an acceptable state space relate to the idea of a comfort zone it seems the acceptable state space for an organism is necessary for it to persist whereas a comfort zone seems to be a less necessary state repertoire which if exceeded may only indicate to an organism that it's at risk of leaving the acceptable state space

interesting

and then the last written question here looking ahead into part two have people been looking in the computational complexity of the active inference model and how far are we in terms of making this computable

and the reference 2019 paper computational resource demands of the

predictive bayesian brain

excellent and important

question to understand like the

analytical computational complexity like

the big o notation does nesting a model

make it two or is it you know log the

number of levels or is it number of

levels to the sum exponent

um and then also in practice like using

active block for ins or pi mdp

what is the actual runtime of different

algorithms

so

um

this is awesome with the questions that

everyone's added to one like we're going

to keep on returning to chapter one just

like we're going to keep on returning to

the ideas but people can fold it

and continue to add to two

and to the appendices which we're gonna

go to next week um so just as a reminder

next week we're going to be

approaching appendix a and b

which are mathematical

and so just a few thoughts on that

before we return back to the ideas in

the book

in appendix a

you can see that all the equations have

already been added

so if anyone like wants to reference

like what does equation a point 44 mean

um and obviously like there's 60

equations and you could get your

phd learning about some sub function of

just one so we're just trying to

familiarize and get a first coat of

paint so for those with different
mathematical backgrounds
if somebody can derive the proof
amazing
let's take some notes on it and if
somebody is just going to be asking
beginner questions that's also very
important so
that will be in next week's questions
and discussions
but let's just look at what ideas people
wrote and then see if there's any ideas
percolating
so
um
having addressed
some specific uncertainties i get with
the questions we can think about like
what are some impressions that people
were having
and like what was exciting ideas
so if um ben or mike want to
speak to this or if anybody else like
wants to share something jf and then
anyone else
yeah
jf then ben or is your hand raised
no my hand was not raised i'm sorry ben
go for it
yeah thanks can you hear me yep
cool uh yeah maybe i should have posed
this as a question actually i just
thought it was it was interesting this
connection between perception and
learning as essentially the same process
unfolding over different
time scales
because on the one hand this makes sense

to me because one example of learning
might be you can imagine an agent move
into a new city
with which they're completely unfamiliar
and the agent could
sit at a fixed point in the city or walk
around the city and just perceive the
city and they would learn over a longer
time scale they would observe certain
regularities or certain occurrences
and clearly that's one form of learning
but then there's maybe another example
where
you know my mathematical background's
fairly weak and so if i were to try and
learn the kind of fundamentals of the
equations that we were just looking at
um no matter how for how long i
kind of perceive those equations
i'm not going to learn them there's
going to be that i have no access to
kind there's no information gained there
it seems like to learn certain things we
need to scaffold perception
in certain ways and so i suppose my
question would be
when somebody teaches us something that
we don't understand is that merely a
kind of scaffolded perception
or is that kind of learning underwritten
by distinct cognitive processing
um completely separate from perception
great question thanks for unpacking that
and to give one
uh thought
let's imagine a ball moving across the
visual field
it seems like a canonical example of

perception we're perceiving the ball but
it could also be understood as parameter
updating or parameter learning
of
the location of the ball
and then
that's one approach but then also you
pointed towards like this deeper sense
of learning referring to like the
underlying structures like in the ball
movement examples it seems like learning
and perception
are the same but then learning might
also refer to something deeper like the
tendencies of how balls move or one
could
as probably many of us have look at
these
symbols
and perceive them visually
but because we wouldn't have learned the
underlying generative model
that gives rise to these pixels we could
continually be surprised
so what is happening when there's
perception and learning
seemingly happening hand in hand
and then other times where somebody's
speaking a language you don't understand
you certainly hear it you may even be
paying attention to it
yet the next phoneme may still
continually surprise you
so it's a really important question
and
then um mike wrote um
yes please go for it mike
yeah uh so

i think

these are straightforward but
foundational concepts that the
dependence of sensory input on action
and vice versa

creates a framework where we can have
feedback and then

the mathematical constructs let us think
about how we model that and simulate it
thanks for this

also essential point in active inference
there isn't a perception and inference
module that passes the perceptions to
the learning module and the learning
module passes it to the decision making
module the decision making modules of
the action model the inference is
conditioned upon policy selection and
action

and that is going to approach this nexus
of perception cognition and action
in an integrated way that not
that may differ from other frameworks
which was also to the point of that
um

previous question

so

this was um an awesome
session everyone really
appreciated it

i'll upload the video recording

bala following the meeting

next week at the same time we'll be
looking at questions and ideas from
appendix a and b

so just approach those however you'd
like and annotate the coda
and

uh also if anybody wants to like specify
another time when they just want to
discuss or co-work
that can be added
or
in the actin flap discord people can
like join it we can have like a voice
like a study hour or just a time to talk
about stuff or write down the questions
we have so
awesome work everybody thanks a lot